

Lab 4

Part 1

First install the **HSQLDB** database (See **install databases** document)

Modify the given **Lesson4SpringJPADemo** application so that the application stores students in the database

A Student contains the attributes name, phoneNumber and email.

A student has also an address.

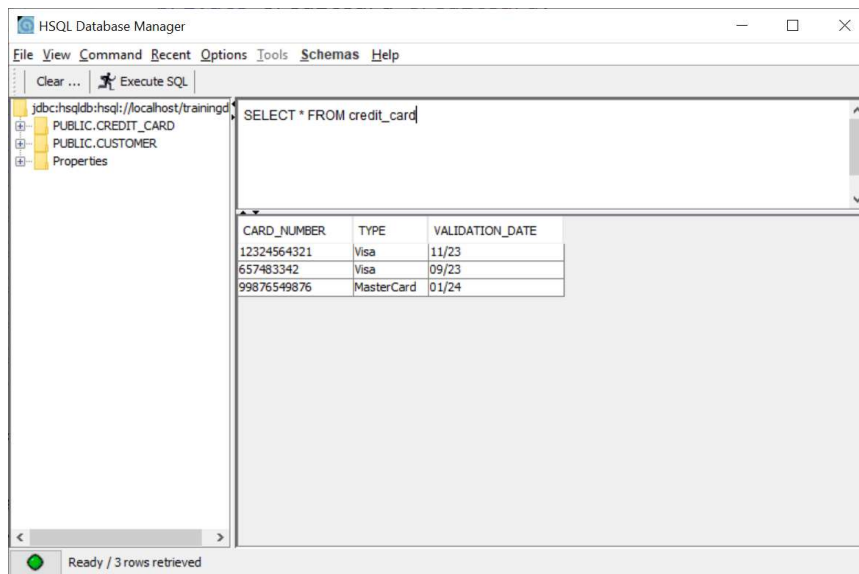
Create a class Address with the attributes street, city and zip.

In the application add 5 students in the database.

Then perform the following queries:

- Get all students
- Get all students with a certain name
- Get a student with a certain phoneNumber
- Get all students from a certain city

Check the data in the database with the HSQLDatabase manager



Part 2

First install **MongoDB** and **MongoCompass** (See install databases document)

Modify the given **Lesson4SpringMongoDemo** application so that the application stores students in the database

A Student contains the attributes name, phoneNumber and email.

A student has also an address.

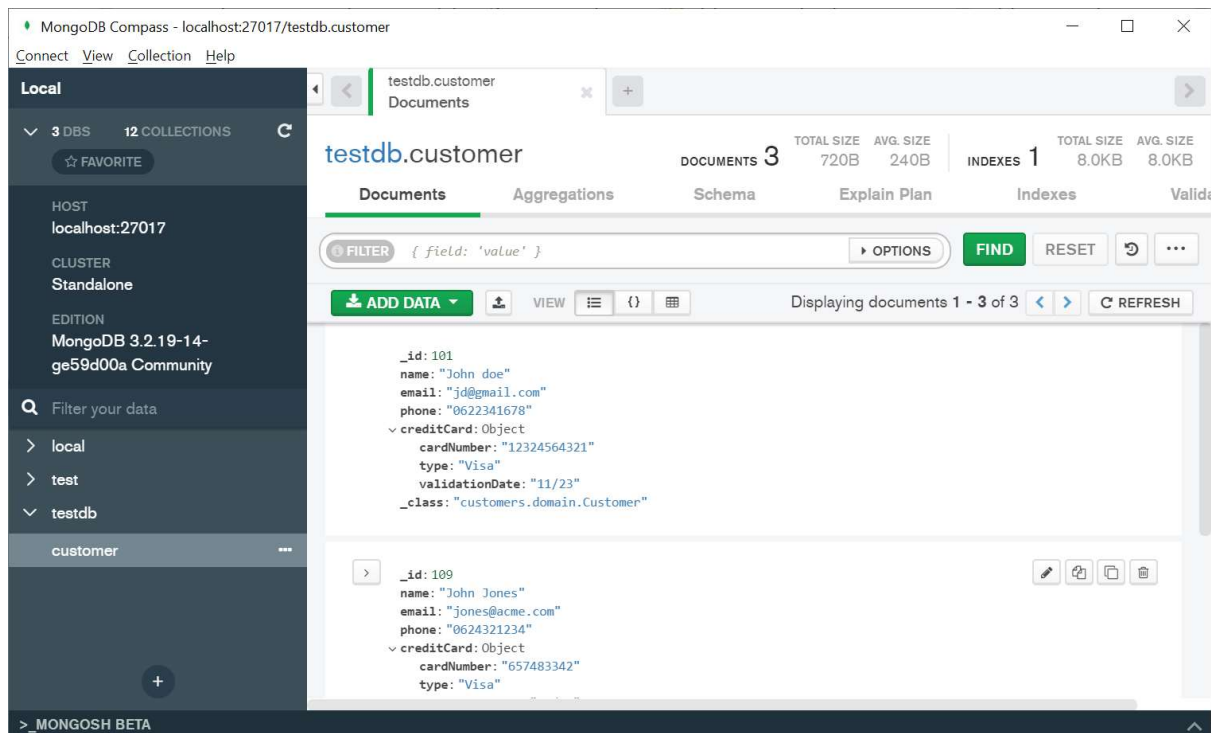
Create a class Address with the attributes street, city and zip.

In the application add 5 students in the database.

Then perform the following queries:

- Get all students
- Get all students with a certain name
- Get a student with a certain phoneNumber
- Get all students from a certain city

Check the data in the collection using Mongo compass.



Part 3

Suppose you need to store the following order in the database:

Ordernumber:122435

Orderdate 11/09/2021

Customer name: Frank Brown

Customer email: fbrown@gmail.com

Customer phone: 0623156543

Total price : 5160.00

quantity	Product number	Product name	price
2	A546	IPhone 12	980.00
4	S333	Samsung Galaxy 12S	800.00

1. Draw the tables including data that you need to store this order in a relational database.
2. Draw the collections including data that you need to store this order in a mongo database.
3. Draw the tables including data that you need to store this order in a cassandra database.
4. Draw the database structure including data that you need to store this order in a neo4j database.

What to hand in?

1. A zip file of part 1
2. A zip file of part 2
3. A PDF of part 3
4. Write a readme.txt file with the following statement and sign with your name:

I hereby declare that this submission is my own original work and to the best of my knowledge it contains no materials previously published or written by another person. I am aware that submitting solutions that are not my own work will result in an NC of the course.

I am aware that I am not allowed to share solutions with other students.

I am aware that if I submit only parts of this lab that points will be subtracted.

I am aware that if my lab submission does not contain this readme.txt file that I do not get points for this lab.

[your name as signature]